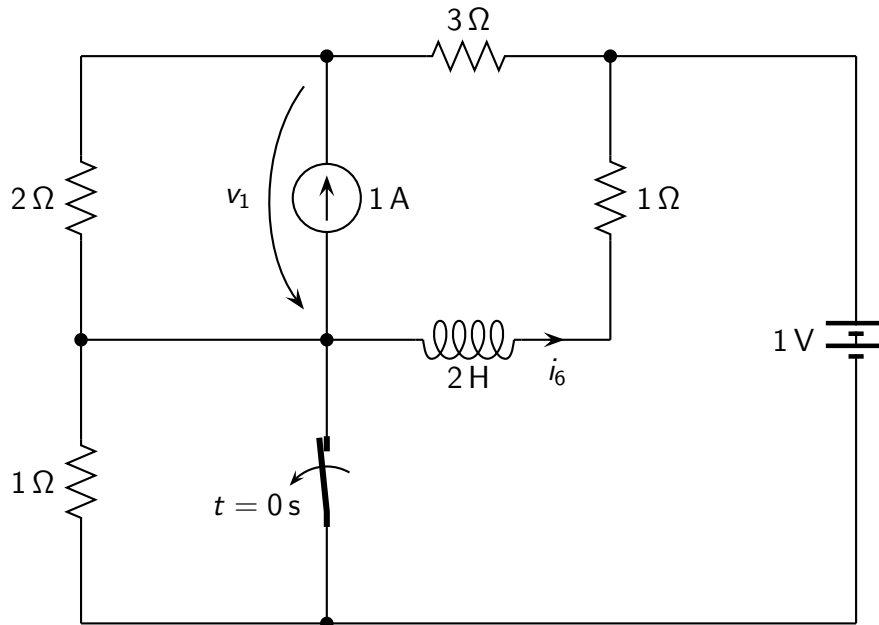


Problem: find the expression of $v_1(t)$, $i_6(t)$ for $t > 0$.



Solution

- Time constant: $\tau = \frac{12}{11} \text{ s}$;
- Initial conditions before switching: $i_6(0^-) = -1 \text{ A}$;
- Initial conditions after switching: $v_1(0^+) = \frac{-4}{3} \text{ V}$; $i_6(0^+) = -1 \text{ A}$;
- Steady-state solution: $v_1(\infty) = \frac{-16}{11} \text{ V}$; $i_6(\infty) = \frac{-7}{11} \text{ A}$;
- Solution for $t > 0$:

$$v_1(t) = \left(\frac{4}{33} e^{-t/\tau} + \frac{-16}{11} \right) \text{ V}$$

$$i_6(t) = \left(\frac{-4}{11} e^{-t/\tau} + \frac{-7}{11} \right) \text{ A}$$